

Weighbridge System — Backend Architecture & Flow

Variables • Databases • APIs • DTOs • Flows • Charts

This document defines a backend blueprint for the weighbridge platform: environment variables, relational schema, REST APIs (with DTOs and status codes), data flows, hardware integration points, and dashboard data derivation. All diagrams are conceptual and suitable for Node/Express + PostgreSQL, but portable to other stacks.

1) Environment & Configuration Variables

Variable	Example / Notes
NODE_ENV	production development
PORT	8080
DATABASE_URL	postgresql://user:pass@host:5432/weighbridge
DB_SCHEMA	public
JWT_SECRET	for auth token signing (HS256)
JWT_EXPIRES_IN	3600s
LOG_LEVEL	info warn error debug
ALLOW_ORIGIN	https://app.example.com (CORS)
ENABLE_SOCKETS	true false (realtime updates)
WEIGHBRIDGE_DEVICE_HOST	192.168.10.50 (Masterdom/PLC gateway)
WEIGHBRIDGE_DEVICE_PORT	502 (Modbus/TCP example)
HARDWARE_POLL_INTERVAL_MS	1000 (status polling cadence)
RATE_LIMIT_WINDOW_MS	60000
RATE_LIMIT_MAX	120 requests/window
STORAGE_PROVIDER	local s3
S3_BUCKET	weighbridge-attachments
S3_REGION	eu-west-1
S3_ACCESS_KEY / S3_SECRET_KEY	Credentials for S3 provider

2) Database Schema (Relational)

Recommended DB: PostgreSQL with a normalized core and JSONB for extensibility.

vehicles

Column	Type	Null?	Notes
id	SERIAL PK	NO	
plate	VARCHAR(20) UNIQUE	NO	Registration number
vehicle_type	VARCHAR(50)	YES	Truck Trailer
model	VARCHAR(100)	YES	
from_another_plant	BOOLEAN	NO	default false
other_plant_name	VARCHAR(100)	YES	
created_at	TIMESTAMP	NO	default now()
updated_at	TIMESTAMP	NO	default now()

drivers

Column	Type	Null?	Notes
id	SERIAL PK	NO	
name	VARCHAR(100)	NO	
phone	VARCHAR(20)	YES	
license_no	VARCHAR(50)	YES	
assigned_vehicle_id	INT FK→vehicles.id	YES	
created_at	TIMESTAMP	NO	default now()
updated_at	TIMESTAMP	NO	default now()

transactions

Column	Type	Null?	Notes
id	SERIAL PK	NO	
document_type	VARCHAR(50)	NO	Delivery Note Invoice
delivery_note_no	VARCHAR(50)	YES	
reference_no	VARCHAR(50)	YES	
batch	VARCHAR(50)	YES	
source_plant	VARCHAR(100)	YES	
destination	VARCHAR(100)	YES	
operation	VARCHAR(100)	NO	Outbound Product Dispatch Inbound Material Receipt
vehicle_id	INT FK→vehicles.id	NO	
material	VARCHAR(100)	YES	
supplier	VARCHAR(100)	YES	
cement_type	VARCHAR(100)	YES	
bag_size	VARCHAR(50)	YES	
w1	DECIMAL(12,2)	YES	
w2	DECIMAL(12,2)	YES	
net_weight	DECIMAL(12,2)	YES	computed: w1 - w2

created_at	TIMESTAMP	NO	default now()
updated_at	TIMESTAMP	NO	default now()

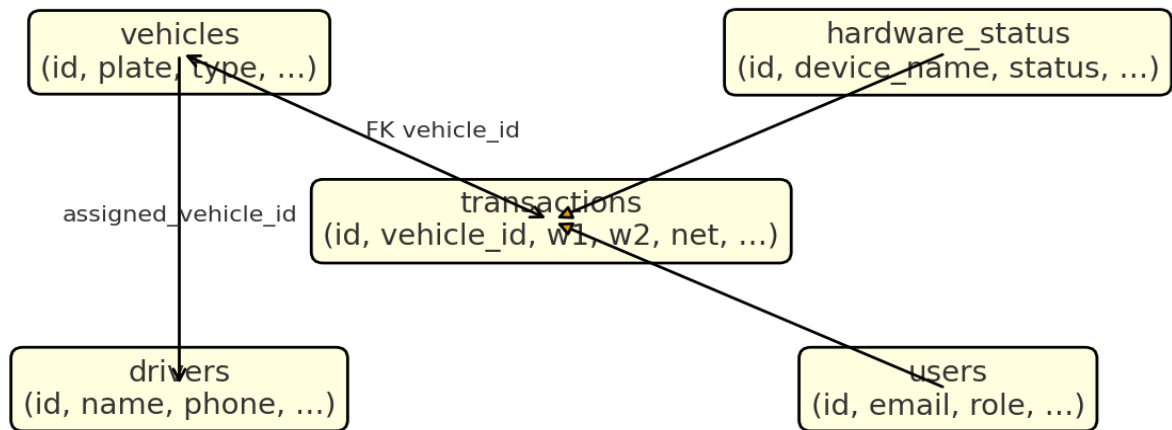
hardware_status

Column	Type	Null?	Notes
id	SERIAL PK	NO	
device_name	VARCHAR(100)	NO	e.g., Vehicle Position Sensor
status	VARCHAR(30)	NO	Healthy Fault Offline
last_heartbeat	TIMESTAMP	YES	
meta	JSONB	YES	port, address, measuring_range, etc
created_at	TIMESTAMP	NO	default now()
updated_at	TIMESTAMP	NO	default now()

users

Column	Type	Null?	Notes
id	SERIAL PK	NO	
email	VARCHAR(120) UNIQUE	NO	
password_hash	VARCHAR(255)	NO	bcrypt
role	VARCHAR(30)	NO	admin operator viewer
created_at	TIMESTAMP	NO	default now()
updated_at	TIMESTAMP	NO	default now()

3) Entity-Relationship Diagram (Simplified)



4) REST API Endpoints & DTOs

Auth

Endpoint	Notes
POST /api/auth/login	200 {token, user} • 401 invalid creds
POST /api/auth/register	201 user • 403 if not admin
GET /api/auth/me	200 user

Request (login) { "email": "op@example.com", "password": "secret" } Response { "token": "jwt...", "user": { "id": 1, "email": "op@example.com", "role": "operator" } }

Vehicles

Endpoint	Notes
GET /api/vehicles	200 list, supports ?q=&page=&size=
POST /api/vehicles	201 created • 409 duplicate plate
PUT /api/vehicles/:id	200 updated • 404 not found
DELETE /api/vehicles/:id	204 no content

DTO { "plate": "KDC123A", "vehicle_type": "Truck", "model": "Isuzu", "from_another_plant": false, "other_plant_name": null }

Drivers

Endpoint	Notes
GET /api/drivers	200 list
POST /api/drivers	201 created
PUT /api/drivers/:id	200 updated
DELETE /api/drivers/:id	204 no content

DTO { "name": "John Doe", "phone": "+254700000000", "license_no": "DL-12345", "assigned_vehicle_id": 12 }

Transactions

Endpoint	Notes
GET /api/transactions	200 list; filters plate, dateFrom, dateTo, op
GET /api/transactions/:id	200 item • 404 not found
POST /api/transactions	201 created (captures w1)
PUT /api/transactions/:id	200 updated (e.g., add w2 & net)
DELETE /api/transactions/:id	204 no content

Create (w1) { "document_type": "Delivery Note", "delivery_note_no": "DN-001", "operation": "Outbound Product Dispatch", "vehicle_id": 12, "material": "Cement", "supplier": "Acme", "w1": 32620.00 } Update (w2) { "w2": 10980.00 // server computes net_weight = w1 - w2 }

Hardware

Endpoint	Notes
GET /api/hardware/status	200 list of device statuses
POST /api/hardware/status	201 created/updated device status
GET /api/hardware/health	200 {healthy: boolean, details: {...}}

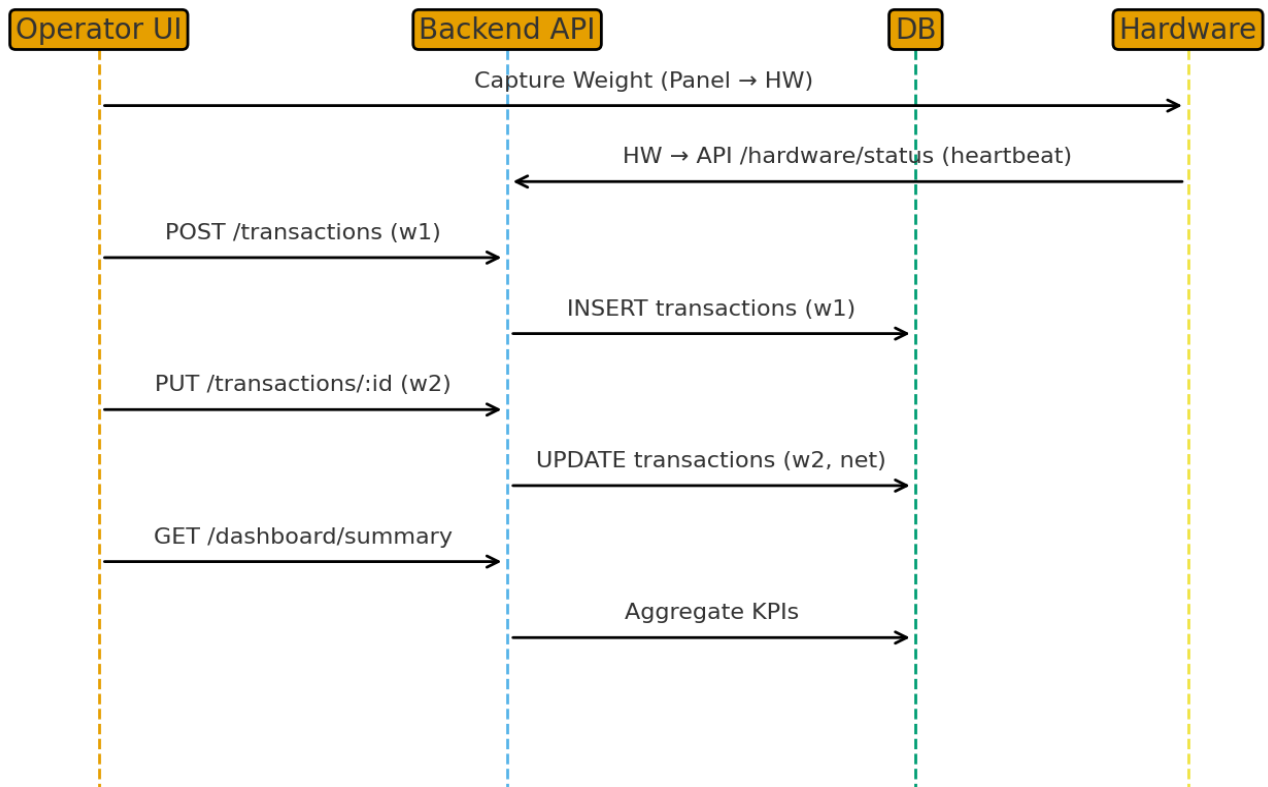
Device Status { "device_name": "Gate Controller", "status": "Healthy", // Healthy | Fault | Offline "meta": { "ip": "192.168.10.60" } }

Dashboard

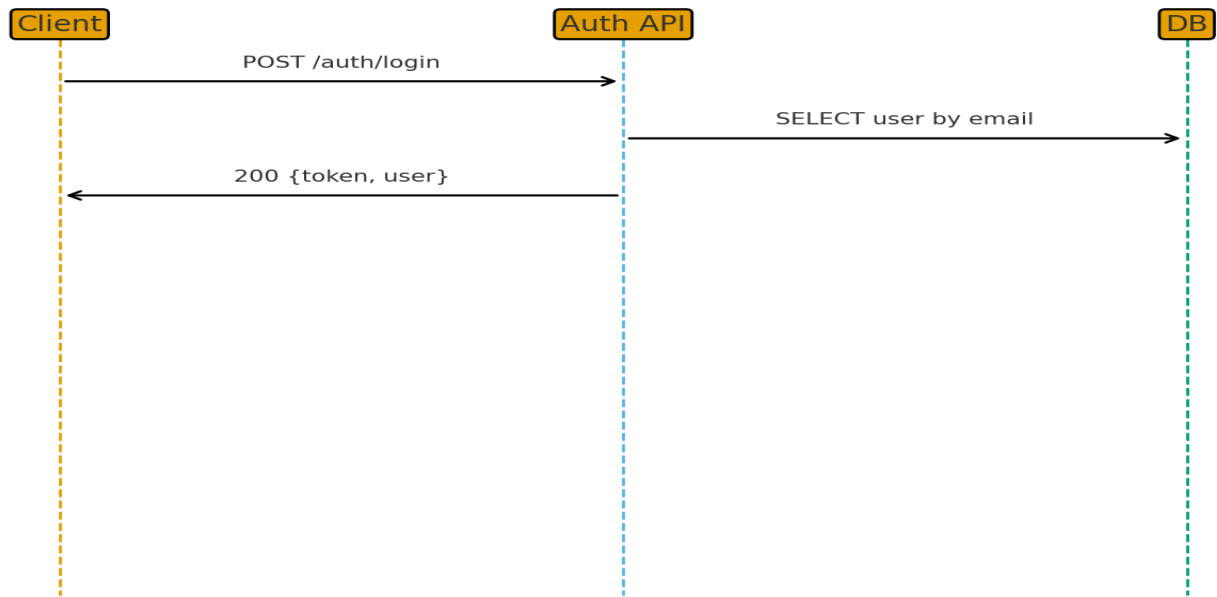
Endpoint	Notes
GET /api/dashboard/summary	200 KPIs
GET /api/dashboard/recent?limit=10	200 latest transactions
GET /api/dashboard/weights/daily	200 time series

Summary Response { "totalTransactions": 120, "inbound": 45, "outbound": 75, "avgNetWeight": 10350.5, "activeVehicles": 23, "pendingSecondWeights": 8 }

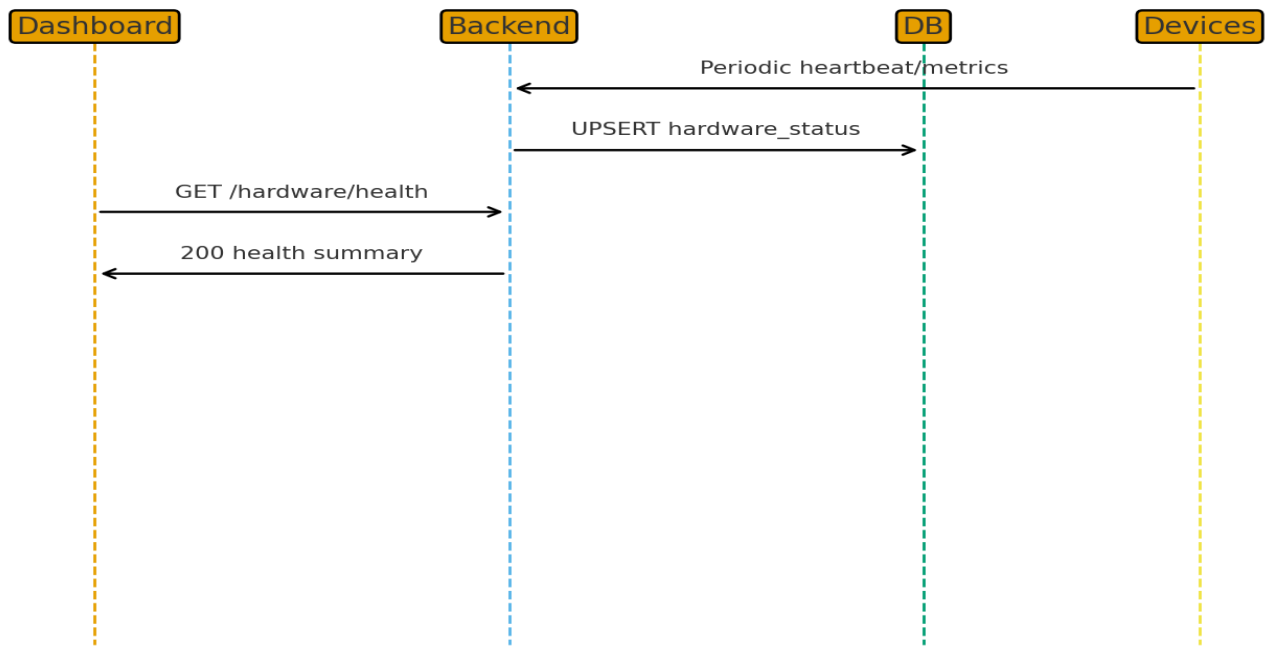
5) Sequence — Weighing Lifecycle



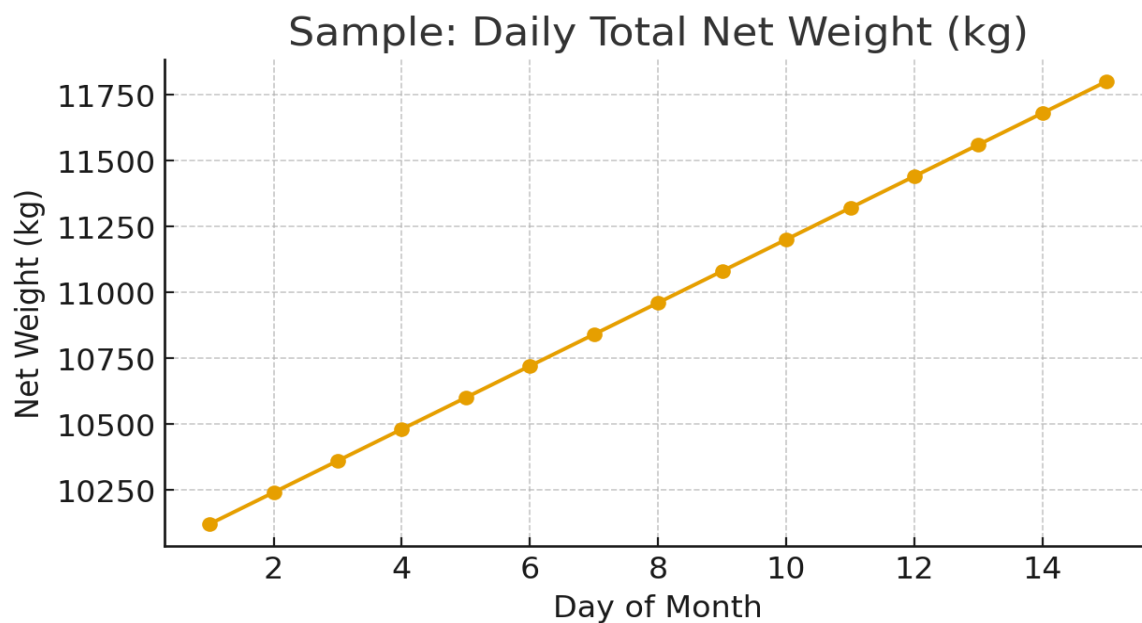
6) Sequence — Authentication



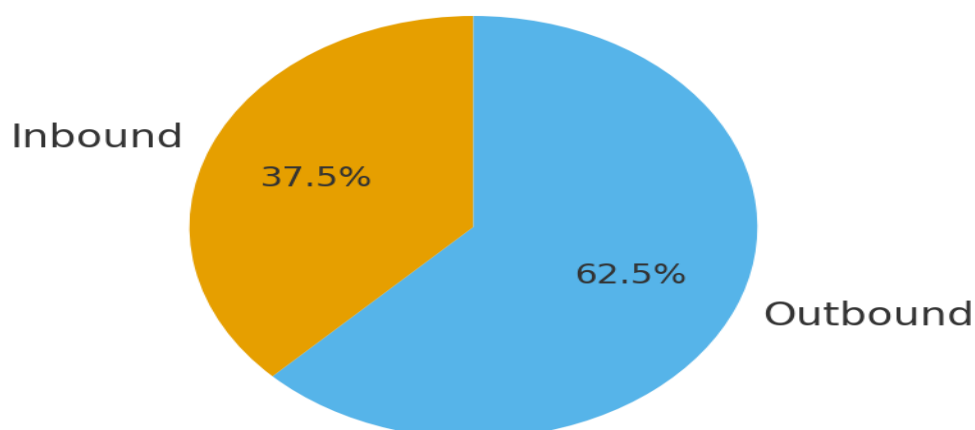
7) Sequence — Hardware Health & Monitoring



8) Dashboard KPIs & Charts (Examples)



Sample: Inbound vs Outbound



9) API Conventions — Errors, Pagination, Security

Errors (JSON)

```
{ "error": "VALIDATION_ERROR", "message": "plate is required", "details": { "plate": "must be unique" } }
```

Pagination

Query: ?page=1&size=20 Response Headers: X-Total-Count Response Body: items[], page, size

Security

- JWT (Authorization: Bearer <token>) - RBAC: admin/operator/viewer - Rate limiting & audit logs - CORS allow-list - Input validation & sanitization

10) Example SQL DDL (Excerpt)

```
CREATE TABLE vehicles ( id SERIAL PRIMARY KEY, plate VARCHAR(20) UNIQUE NOT NULL, vehicle_type VARCHAR(50),
model VARCHAR(100), from_another_plant BOOLEAN NOT NULL DEFAULT FALSE, other_plant_name VARCHAR(100),
created_at TIMESTAMP NOT NULL DEFAULT NOW(), updated_at TIMESTAMP NOT NULL DEFAULT NOW() ); CREATE
TABLE transactions ( id SERIAL PRIMARY KEY, document_type VARCHAR(50) NOT NULL, delivery_note_no VARCHAR(50),
reference_no VARCHAR(50), batch VARCHAR(50), source_plant VARCHAR(100), destination VARCHAR(100), operation
VARCHAR(100) NOT NULL, vehicle_id INT NOT NULL REFERENCES vehicles(id), material VARCHAR(100), supplier
VARCHAR(100), cement_type VARCHAR(100), bag_size VARCHAR(50), w1 DECIMAL(12,2), w2 DECIMAL(12,2), net_weight
DECIMAL(12,2), created_at TIMESTAMP NOT NULL DEFAULT NOW(), updated_at TIMESTAMP NOT NULL DEFAULT
NOW() );
```